

HAT-P-37b

R-2613

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-37_250711 · created 2026-08-28T01:42:00+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460867.7851 +/- 0.0091 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.171 +/- 0.043
Transit depth (Rp/Rs)^2	2.91 +/- 1.46 [%]
Orbital Inclination (inc)	89.67 +/- 2.0
Airmass coefficient 1 (a1)	1.13 +/- 0.038
Airmass coefficient 2 (a2)	-0.11 +/- 0.11
Scatter in the residuals of the lightcurve fit is	3.38 %
Best Comparison Star	#1 - [225, 425]
Optimal Method	PSF photometry
Transit Duration (day)	0.0467 +/- 0.0096

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.171 ± 0.043 vs 0.1394 (deviation 0.6σ)
Duration fit vs published	0.0467 d vs 0.0983 d (deviation 4.41σ)
Mid-time O-C	28.3 min
Depth SNR	3.3
Residual scatter	3.38 %

Light curve

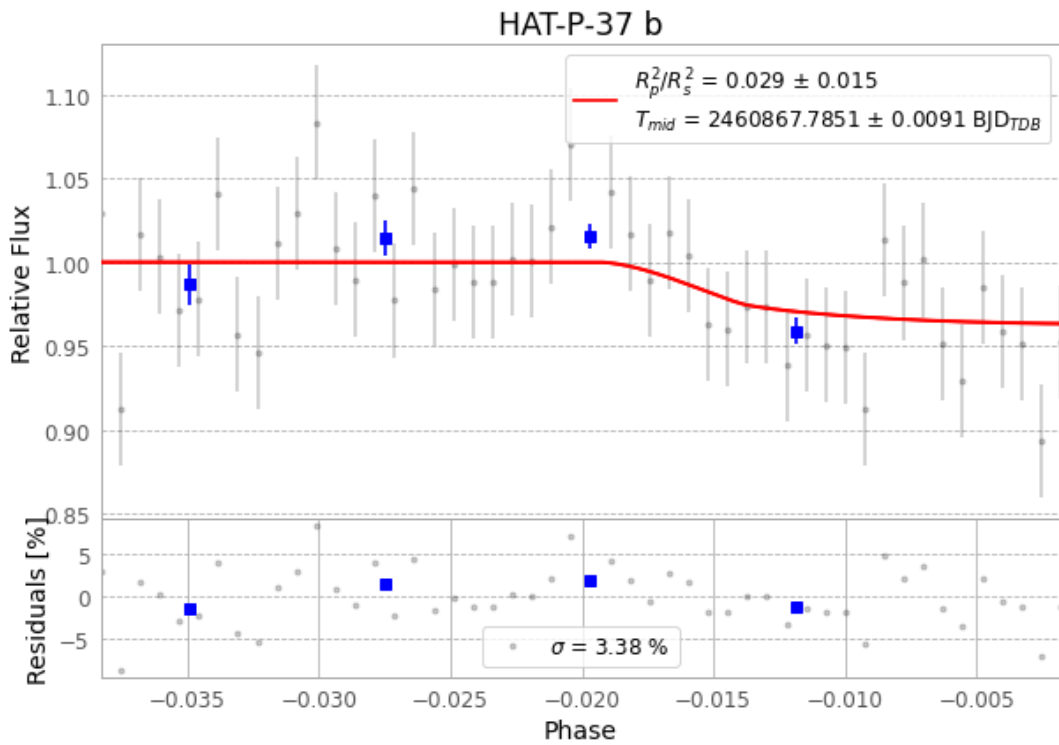


Figure 1 — FinalLightCurve_HAT-P-37 b_2025-07-10.png (SHA-256 a9c0496174faf7c1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [349, 208], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 15177da330dc7a4c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

50 FITS frames (33 MB), HATP-37250711041215.FITS → HATP-37250711063915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-37-250711.log (07a32d314237...), FinalLightCurve_HAT-P-37 b_2025-07-10.png (a9c0496174fa...), FinalLightCurve_HAT-P-37 b_2025-07-10.csv (03f784e0839a...)

Compute resources

Wall time 184.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-28 01:42Z.